

MORCHED AMMAR

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PROFESSIONAL SUMMARY

As a junior developer with a strong background in C++ and Python, I have developed a solid foundation in firmware and embedded AI engineering. With experience in interrupt-driven STM32 firmware and FreeRTOS multitasking on ESP32, I am confident in my ability to design and implement efficient systems. My proficiency in C/C++, Python, and various embedded protocols such as CAN, OBD-II, MQTT, I2C, SPI, and BLE enables me to tackle complex projects. I am excited to apply my skills and knowledge to contribute to innovative projects and continue growing as a developer.

TECHNICAL SKILLS

C++, Python, C, CAN, OBD-II, MQTT, I2C, SPI, BLE, JavaScript, ESP8266, ESP32, Raspberry Pi, HTTP, REST, Sensor-based architectures, YOLOv8, OpenCV, ONNX, Edge AI, Reinforcement Learning, UART, Wi-Fi, TCP/IP, PCB design, EasyEDA, Altium, Sensor integration, Actuator control, Git, STM32CubeIDE, Arduino IDE, VS Code, Flutter, Dart, Flask

PROFESSIONAL EXPERIENCE

Capgemini- Final year intern: mobile Dev

Feb 2025/Aout 2024

Capgemini

- Built a Flutter/Dart OBD-II client that streamed live data from real ECUs, demonstrating my ability to work with automotive diagnostics and real-time data processing.
- Designed and implemented a Flask-based ECU simulator, showcasing my understanding of embedded systems and protocol interactions.
- Developed and tested a Wi-Fi/TCP-IP telemetry extraction pipeline, highlighting my capacity to integrate multiple technologies and protocols.

Sofrecom-Summer internship: embedded system

July 2024/ August 2024

Sofrecom

- Developed a REST API that leveraged the 51Degrees device detection engine, demonstrating my ability to work with APIs and device interactions.
- Built a Raspberry Pi backend that processed sustainability profiles and transmitted data to an ESP8266, showcasing my understanding of IoT and embedded systems.

PROJECTS

TrafficOS — Deep RL Intersection Controller

Technologies: OpenAI Gymnasium, PPO, Stable Baselines3, Raspberry Pi 5, ONNX

Engineered a custom OpenAI Gymnasium environment modeling a 4-way intersection with an 8-dimensional observation space (per-lane vehicle density + emergency vehicle priority flags). Trained a PPO agent (Stable Baselines3) over 500,000 timesteps, achieving a 30–35% reduction in average vehicle wait time vs. fixed-timer controllers.

Padel Player Tracking — Computer Vision & IoT

Technologies: STM32F407VG, I2C, SPI, MQTT, Wi-Fi

Engineered modular embedded firmware in C on STM32F407VG managing 3 sensors (LEM LTS 6-NP, DHT11, PIR) across I2C and SPI buses with a layered driver architecture enabling sensor hot-swap without core firmware changes. Implemented MQTT-over-Wi-Fi cloud pipeline for continuous telemetry upload, enabling remote monitoring and real-time actuation commands from a cloud dashboard.

IoT Energy Monitoring System — STM32F407VG

Technologies: STM32F407VG, I2C, SPI, MQTT, Wi-Fi

Engineered modular embedded firmware in C on STM32F407VG managing 3 sensors (LEM LTS 6-NP, DHT11, PIR) across I2C and SPI buses with a layered driver architecture enabling sensor hot-swap without core firmware changes. Implemented MQTT-over-Wi-Fi cloud pipeline for continuous telemetry upload, enabling remote monitoring and real-time actuation commands from a cloud dashboard.

EDUCATION

Bachelor's degree in electronics and automation — Higher Institute of Industrial Management (2018-2022)

Bachelor's degree in computer engineering — Private higher school of engineering and technology (Sept 2022 - Oct 2025)

CERTIFICATIONS

- Nvidia: Applications of AI for Anomaly Detection (Nov2024)
- Nvidia: Applications of AI for Predictive Maintenance (Nov2024)

LANGUAGES

English (B2), French (B2)